Troy Mayer

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EDUCATION

Purdue University, School of Mechanical Engineering and School of Science — **West Lafayette, IN**
B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics — **May 2028**
Clubs: Club Swimming, House Band — **GPA 3.32/4**

PROFESSIONAL EXPERIENCE

Boston Dynamics — **Waltham, MA**
Sensors and Perception Co-Op – Atlas — **July 2026 – Present**

- Mapping impacts onto the robot's mesh with encoders and forward kinematics, estimating their energy from Khatib's operational-space effective mass and an OLS velocity fit, and building a physics-based MuJoCo impact simulation.
- Building an impact forensics pipeline and Flask/Plotly dashboard on a Linux VM covering ~28k impact events from ~900 robot test logs, with ETL from a time-series telemetry store into SQLite, a 3D impact-location heatmap, and per-event waveform inspection used by engineers during testing.
- Modeling hardware thermal behavior in Python, using Savitzky-Golay filtering and multi-start least squares to fit heating curves and predict steady-state temperatures across hundreds of units without full-length tests.
- Designing CATIA fixtures, mounts, and shear features, using pin-shear failure analysis to optimize geometry.

Caterpillar (via Purdue Data Mine) — **West Lafayette, IN**
Machine Learning Researcher — **January 2026 – May 2026**

- Built LSTM and XGBoost models forecasting next-day primer, topcoat, and solvent consumption from five years of paint-line data, with Monte Carlo confidence intervals on each forecast, achieving a 4.7% reduction in material usage.
- Built Python pipelines to ingest, clean, and consolidate five years of noisy legacy data into one daily series, contributing to a 3.3% reduction in manufacturing cycle time.

Chortos Lab – Soft Adaptive Systems and Structures — **West Lafayette, IN**
Undergraduate Researcher — **August 2025 – May 2026**

- Built data acquisition systems in LabVIEW and MATLAB, processing and normalizing datasets of 50,000+ points to quantify sub-millimeter material behavior.
- Wrote a Python computer-vision tracker resolving cantilever displacement to 0.01 mm, guiding gel formulations that increased force output by over 50%.

PROJECTS

Purdue ASME – Biomechanics — **West Lafayette, IN**
Active Orthosis – Electronics Team Lead — **April 2024 – May 2026**

- Designed and implemented a 1 kHz PI velocity controller in C++ on an ESP32-S3, tuned from a first-order motor model with feed-forward and anti-windup, and protected by 10 fault monitors, a hardware watchdog, and a kill switch.
- Cut the delay from a muscle signal (EMG) to arm movement from 103 ms to 43 ms using a Bayesian signal estimator, tested in simulation on real recorded EMG data.
- Led a 5-engineer team and redesigned the 4-channel EMG board in KiCad with an INA333 front end, right-leg drive, and ISO7763 motor isolation, raising day-to-day gesture decoding from 31% to 87% in emulated tests.

Asteroid Hopping Robot – Independent Research
Purdue University & Universidad Carlos III de Madrid — **October 2026 – Present**

- Building a three-legged hopping robot that changes leg shape between hops, and a pendulum rig to test landings.
- Designing the leg motor control that pushes off for each hop and optimizing it to hit a target launch speed, plus knee impedance control for landing and reaction-wheel control to reorient mid-air.
- Simulating it in MuJoCo to size each hop, comparing risk-limited rules that trade distance against the chance of escaping the asteroid or tipping over.

Legged Robot State Estimation and Control
Personal Project — **August 2026 – October 2026**

- Built a Kalman filter in C++ that estimates a simulated four-legged robot's velocity from its IMU and the feet on the ground (0.013 m/s error vs. 1.38 m/s IMU-only), plus walking and QP balance controllers that run on it.

SKILLS & INTERESTS

- **Tools:** C++, Python, MATLAB, SQL, LabVIEW, CAD (CATIA, NX, Fusion, Creo), FEA Analysis, KiCad, Snowflake, Git, Linux, Eigen, MuJoCo
- **Languages:** English (Native), Turkish (Fluent), Spanish (Fluent - New Jersey Seal of Biliteracy).
- **Hobbies:** Lifting, Hiking, Guitar, Saxophone